



## F329 Deci-Newton Loadcell

The F329 is a specialist force measurement device.

Due to its finely balanced strain system and delicate structure there are some restrictions with respect to its use that need to be considered:

- The loadcell's self mass will influence the zero load output upon changes of orientation and with acceleration therefore dynamic applications are limited.
- The low stiffness characteristic requires the force system to provide enough deflection for the equilibrium deflection to be achieved.

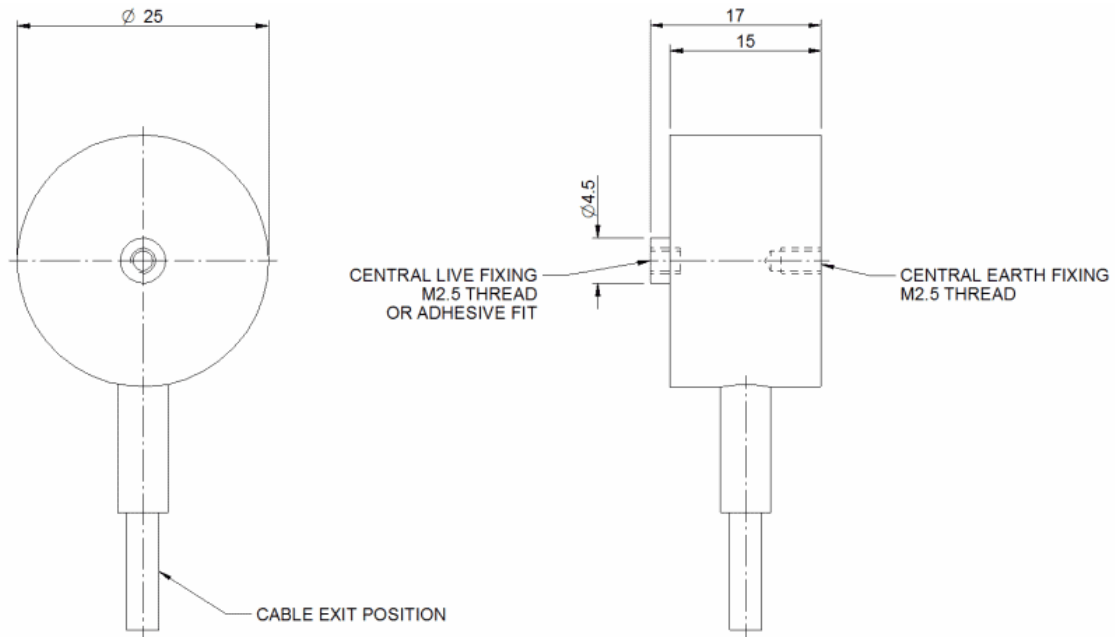
The robust strain system allows judicious application of probes or structure fixings. Input torque to the live force input boss must be minimal, normal frictional torques achieving thread insertion should not be exceeded i.e. thread locking by adhesive is suggested. Additional information can be found in [Engineering Sheet E038](#).

We are happy to design variants of this loadcell to meet your specific requirements. Please consult our engineering department.

### Features

- ✓ Low force high resolution loadcell
- ✓ Best resolution 4 microNewton
- ✓ Passive Wheatstone bridge
- ✓ Foil strain gauges
- ✓ High thermal stability
- ✓ Durable design
- ✓ Translational deflection

## Outline



## Specifications

| Parameter                                         | Value                 | Unit |
|---------------------------------------------------|-----------------------|------|
| Non-linearity - Terminal                          | ±0.2                  | % RL |
| Hysteresis                                        | ±0.1                  | % RL |
| Creep - 20 minutes                                | ±0.2                  | % AL |
| Repeatability                                     | ±0.025                | % RL |
| Rated output - Nominal                            | 0.35 to 0.55          | mV/V |
| Zero load output (loadcell orientated vertically) | ±10                   | % RL |
| Temperature effect on rated output per °C         | ±0.005                | % AL |
| Temperature effect on zero load output per °C     | ±0.029                | % RL |
| Temperature range - Compensated                   | -10 to +50            | °C   |
| Temperature range - Safe                          | -10 to +80            | °C   |
| Excitation voltage - Recommended                  | 5                     | V    |
| Excitation voltage - Maximum                      | 10                    | V    |
| Bridge resistance                                 | 2500                  | Ω    |
| Insulation resistance - Minimum at 50Vdc          | 500                   | MΩ   |
| Mechanical stiffness                              | 1.3 x 10 <sup>3</sup> | N/m  |
| Overload - Safe                                   | 50                    | % RL |
| Live mass                                         | 0.5                   | g    |

## Order Codes

| Code       | Description                    |
|------------|--------------------------------|
| F329CF00A0 | Compression, unrationalised    |
| F329TF00A0 | Tension, unrationalised        |
| F329UF00A0 | Bi-directional, unrationalised |

## Notes

- AL = Applied load.
- RL = Rated load.
- Temperature coefficients apply over the compensated range.
- The load must be applied directly through the central loading axis.
- Tare mass is limited to 25% RL, higher values are possible by prior arrangement with our engineering department.
- Orientation effects will create errors if the loadcell changes orientation during the measurement process.
- Due to the very low stiffness we advise you to consult our engineering department to discuss the viability of your application for this loadcell.

## Connections

The loadcell is fitted with 2 metres of PVC insulated 4 core screened cable type 7-1-4C.

Excitation + = Red, Excitation - = Blue, Signal + = Yellow, Signal - = Green, Screen = Orange.

Reverse the signal connections to obtain a positive signal in tension mode. The screen is not connected to the loadcell body.

This loadcell has compensation components housed in a capsule located in the loadcell cable 100mm from the free end. Capsule dimensions are Ø10mm by 57mm.